

WanderPlan

Product Requirements Document

AI-Powered Conversational Travel Planning Platform

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Table of Contents

1. Executive Summary
2. Product Vision & Goals
3. Target Audience
4. Core Features
5. User Experience & Interface
6. Technical Architecture
7. AI & Machine Learning
8. Success Metrics
9. Roadmap & Milestones
10. Risks & Mitigation

1. Executive Summary

WanderPlan is an AI-powered travel planning platform that revolutionizes trip planning through natural, conversational interactions. Unlike traditional booking platforms with complex forms, WanderPlan introduces **Any**a — an intelligent AI assistant that guides users through personalized itinerary creation using voice and text conversations.

Key Differentiators:

- **Conversational Interface:** Replace multi-step wizards with natural dialogue
- **Voice-First Design:** Fully functional voice input and output with Indian English accent
- **AI-Powered Personalization:** Leverages Gemini 2.0 Flash for context-aware recommendations
- **Community Intelligence:** Incorporates real traveler experiences from Reddit and Wikivoyage
- **Transparent Pricing:** No hidden fees, budget-aware planning from the start

2. Product Vision & Goals

2.1 Vision Statement

"Make travel planning as simple as having a conversation with a knowledgeable friend."

2.2 Product Goals

- **Reduce planning time:** From 5-10 hours to under 15 minutes for a complete itinerary
- **Increase accessibility:** Enable voice-first planning for users uncomfortable with complex interfaces
- **Improve personalization:** Deliver itineraries that match user preferences with 90%+ satisfaction
- **Build trust:** Transparent pricing, no spam, community-validated recommendations

3. Target Audience

Primary: Young Professionals (25-35)

Time-constrained individuals seeking quick, personalized trip planning. Comfortable with technology, prefer voice/chat over forms. Budget: ■50K-2.5L per trip.

Secondary: Families (30-45)

Parents planning kid-friendly vacations. Need detailed itineraries with safety considerations, age-appropriate activities, and flexible pacing. Budget: ■1L-5L per trip.

Tertiary: Senior Travelers (55+)

Retirees with time and budget for leisure travel. Prefer voice interaction, need clear guidance, value relaxed pace. Budget: ■2L-10L+ per trip.

4. Core Features

4.1 Anya - AI Travel Assistant

Anya is the conversational interface that replaces traditional forms with natural dialogue. She guides users through 9 key inputs (purpose, origin, destination, dates, group, budget, accommodation, pace, themes) via text or voice.

- **Voice Mode:** Single animated button for continuous voice interaction (speak + listen)
- **Smart City Suggestions:** Recommends destinations based on user preferences when exploring
- **Multi-Select Themes:** Culture, Food, Adventure, Nature, Photography, Shopping, etc.
- **Refinement Loop:** 'Anything else before I generate?' step for final tweaks
- **Persona:** Female, aged 20-25, Indian English accent, friendly and energetic tone

4.2 Itinerary Generation

AI-generated day-by-day schedules with timestamps, cost estimates, locations, and activity descriptions. Powered by Gemini 2.0 Flash with real-time streaming (30-60 seconds for 5-7 day trips).

- **Timeline View:** Visual day-by-day cards with activities, duration, and costs
- **Interactive Map:** Leaflet integration with pins for all activity locations
- **YouTube Embeds:** Video previews for attractions and activities
- **Transit Warnings:** Alerts when travel time between activities exceeds 30 minutes
- **Expense Breakdown:** Category-wise cost summary (Transport, Food, Activities, etc.)

4.3 Booking Integration

Seamless handoff to trusted booking platforms for hotels, flights, and car rentals.

- **Hotels:** Booking.com, Agoda, Hotels.com
- **Flights:** Skyscanner, Google Flights
- **Car Rentals:** Rentalcars.com

4.4 Travel Intelligence

- **Best Time to Visit:** Weather data, temperature, rainfall, tourist density analysis

- **Community Tips:** AI-curated insights from Reddit and Wikivoyage (6 tips per destination)
- **Budget Recommendations:** Data-driven budget guidance based on destination and group size

5. User Experience & Interface

5.1 Desktop Layout (Primary)

Three-column responsive design optimized for 1920×1080 and above.

- **Left Column (20%):** Trip metrics, booking CTAs, expense breakdown, currency selector
- **Center Column (55%):** Itinerary timeline or comparison panel
- **Right Column (25%):** Map, best time widget, travel tips
- **Overlay:** Conversational wizard (Anya) opens by default on first visit

5.2 Conversational Flow

1. User opens app → Anya greets: "Hi! I'm Anya from WanderPlan..."
2. Sequential prompts for 9 inputs (purpose → origin → destination → dates → group → budget → accommodation → pace → themes)
3. City suggestions for exploring/country modes via Gemini
4. Refinement step: "Anything else before I generate?"
5. Trip summary card with edit options
6. Generate itinerary (30-60s streaming)
7. Wizard stays open with "View Itinerary" CTA

5.3 Voice Interaction

Single  button with pulsating animation. Click to toggle voice mode — Anya both speaks and listens in a continuous loop. Auto-restart after each response.

- **Recognition:** Web Speech API, language: en-IN
- **Synthesis:** Browser Speech Synthesis, priority: Indian English female voices
- **Voice Characteristics:** Pitch 1.15 (young female), rate 1.05 (energetic), volume 1.0

6. Technical Architecture

6.1 Frontend Stack

- **Framework:** Next.js 16.2.9 with Turbopack
- **Language:** TypeScript 5.x
- **State Management:** Zustand (4 stores: app, tripConfig, itinerary, wizardChat)
- **Styling:** Tailwind CSS v4
- **Maps:** Leaflet with OpenStreetMap tiles
- **Charts:** Recharts
- **Deployment:** Vercel (recommended)

6.2 Backend Stack

- **Framework:** FastAPI (Python 3.9+)
- **LLM:** Google Gemini 2.0 Flash (gemini-2.0-flash-exp)
- **Vector Database:** Qdrant (in-memory mode, 384-dim embeddings)
- **Embedding Model:** sentence-transformers/all-MiniLM-L6-v2
- **Scheduler:** APScheduler (Reddit refresh every 6 hours)
- **Deployment:** Railway / Render (recommended)

6.3 External APIs

- **Geocoding:** Nominatim (OpenStreetMap, rate-limited 1 req/sec)
- **Weather:** Open-Meteo (free, no API key)
- **Community Content:** Reddit JSON API (no auth required)
- **Maps:** OpenStreetMap tiles (free)
- **YouTube:** HTML scraping for thumbnail search

7. AI & Machine Learning

7.1 Gemini 2.0 Flash Integration

Primary LLM for all generative tasks. Chosen for speed (2-5s responses), cost-efficiency (free during preview), and strong JSON structured output.

- **Itinerary Generation:** 30-60s streaming, temperature 0.7, max tokens 4096
- **Chat Refinement (Anya):** 2-5s response, temperature 0.5, structured JSON output
- **City Recommendations:** 3-6s, temperature 0.7, returns 5 cities with descriptions
- **Travel Tips:** 2-4s, temperature 0.8, generates 6 community-style tips (cached)

7.2 Vector Search (Qdrant)

Retrieves relevant community insights from Reddit and Wikivoyage during itinerary generation. Queries use destination + themes → cosine similarity > 0.1 → top 10 results.

- **Embedding Model:** all-MiniLM-L6-v2 (384 dimensions, runs locally)
- **Collections:** reddit_highlights, wikivoyage_content
- **Ingestion:** APScheduler job every 6 hours (Reddit), on-demand (Wikivoyage)
- **Storage:** In-memory (data lost on restart, re-ingestion required)

7.3 Prompt Engineering

All prompts follow structured templates with user profile context (purpose, group, pace, budget, themes) and retrieved community insights. Anya's persona is defined in system prompts with guardrails against non-travel queries.

8. Success Metrics

8.1 User Engagement

- **Wizard Completion Rate:** >80% of users complete all 9 fields
- **Voice Mode Adoption:** >30% of users try voice interaction
- **Average Session Duration:** 12-18 minutes (wizard + itinerary review)
- **Repeat Usage:** >40% of users plan 2+ trips within 6 months

8.2 Product Quality

- **Itinerary Satisfaction:** >90% users rate itinerary 4-5 stars
- **Anya Response Accuracy:** >95% responses correctly interpret user intent
- **Booking Conversion:** >25% of generated itineraries lead to bookings (tracked via affiliate links)
- **Error Rate:** <2% of API calls result in errors

8.3 Performance

- **Itinerary Generation (P95):** <60 seconds for 5-7 day trips
- **Chat Response Time (P95):** <5 seconds
- **Page Load Time (P95):** <2 seconds
- **Uptime:** >99.5%

9. Roadmap & Milestones

9.1 Phase 1: MVP (Current)

- Conversational wizard with Anya
- Voice input and output (Indian English)
- Itinerary generation with Gemini 2.0 Flash
- Booking links integration
- Travel tips and best time widget
- Desktop-optimized UI (mobile warning banner)

9.2 Phase 2: Mobile & Sharing (Q3 2026)

- Responsive mobile web app
- Shareable itinerary links (public/private)
- PDF export with branding
- WhatsApp/email sharing
- Progressive Web App (PWA) support

9.3 Phase 3: Collaboration & Booking (Q4 2026)

- Multi-user trip planning (invite friends/family)
- Real-time collaboration on itineraries
- In-app booking for flights and hotels
- Payment integration (Stripe/Razorpay)
- Booking commission revenue model

9.4 Phase 4: Intelligence & Scale (2027)

- Personalized recommendations based on past trips
- Price tracking and alerts
- Multi-city trip support
- Currency conversion across multiple currencies
- Regional language support (Hindi, Spanish, etc.)

- Integration with Google Calendar/Apple Calendar

10. Risks & Mitigation

Risk	Impact	Mitigation
Gemini API Rate Limits	High	Implement aggressive caching (travel tips already cached), queue requests du
Voice Recognition Accuracy	Medium	Provide text fallback always visible, show transcript for confirmation, improve v
Qdrant Data Loss (In-Memory)	Medium	Migrate to persistent storage (Railway volume), automated backup, graceful d
Reddit API Blocking	Low	Already using Gemini as primary source for tips, Reddit is fallback only
Booking Conversion Low	High	A/B test CTA placement, add urgency messaging (price alerts), improve trust s

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End of Product Requirements Document